



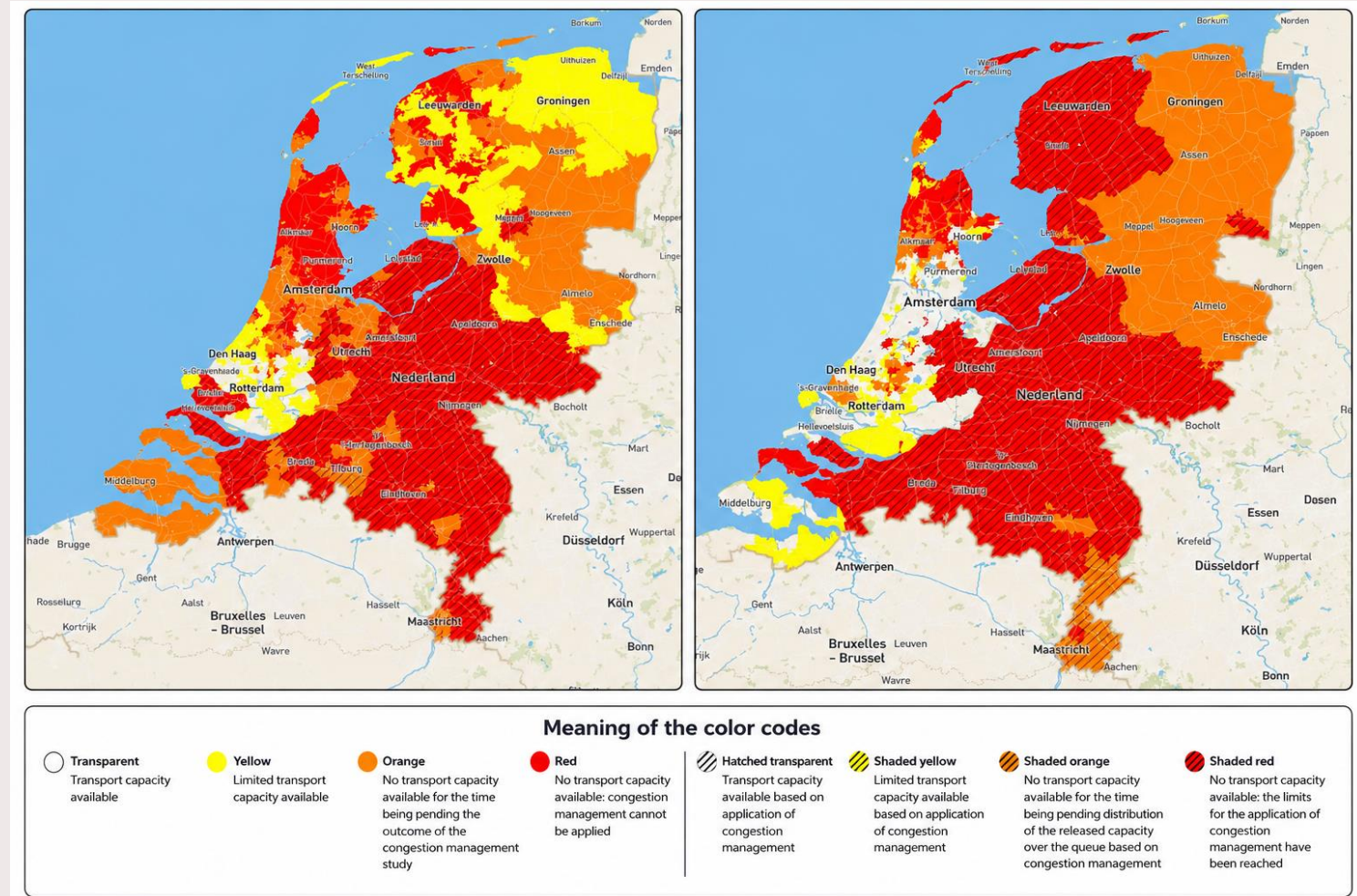
CityLearn Challenge

JUNE 2026

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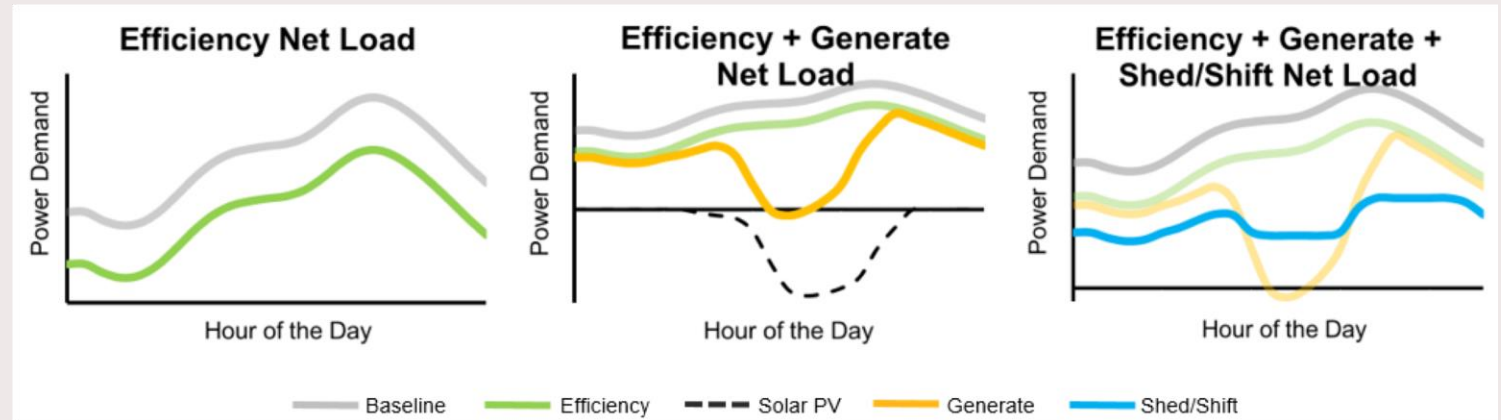
The grid is full

- Net metering (salderingsregeling) phased out → timing of storage & consumption matters
- Netcongestie: DSOs refuse new PV connections
- Demand response beats grid reinforcement: reshape loads, not cables



Buildings must become grid assets

- From passive loads to grid-interactive efficient buildings (GEB)
- Coordinate PV, battery and HVAC to follow grid-level targets
- Single-building flexibility is too small — aggregate at district scale



34%

of energy-related CO₂ from
buildings (2023)

32%

of global energy demand

Research question



Which control architecture best enables a district of residential buildings to track a prescribed daily load profile while maintaining occupant comfort?

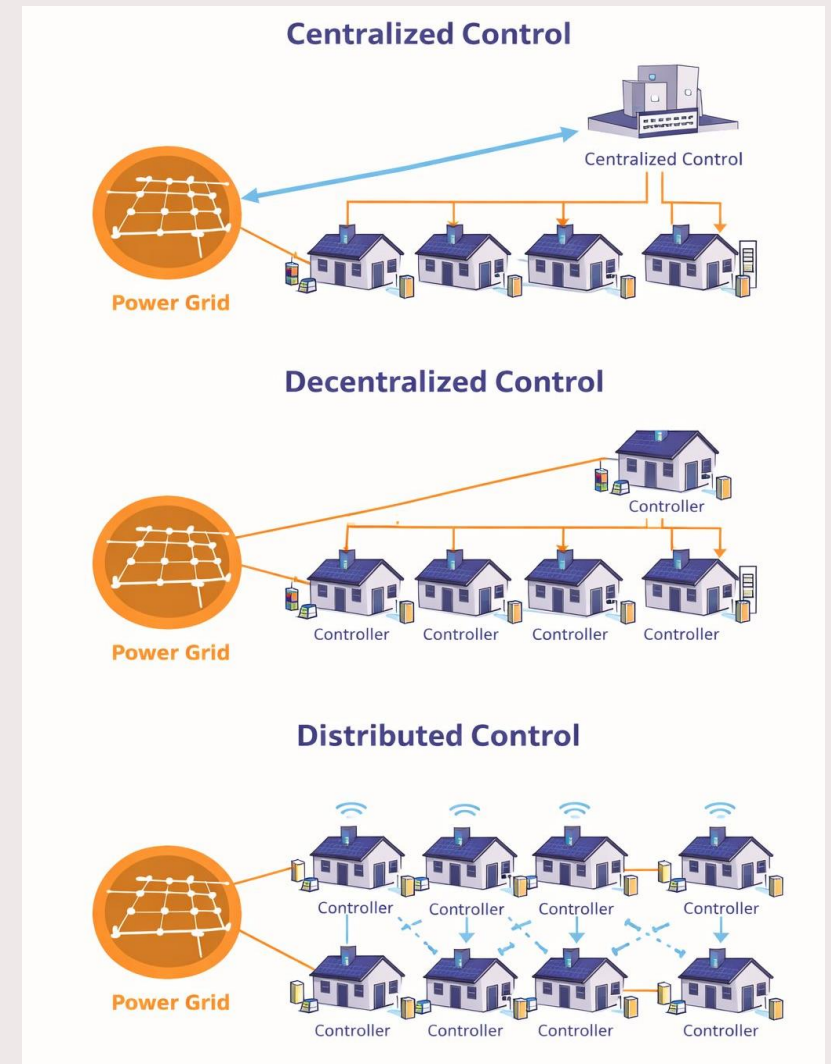
RQ1 · Which coordination works?

RQ2 · How do constraints shape behaviour?

RQ3 · Best tracking–comfort trade-off?

The Testbed: CityLearn, Texas District

- A widely used benchmark environment for developing and evaluating energy management strategies in smart buildings and energy-flexible communities.
- **25 residential buildings**, each equipped with energy assets such as solar panels (PV), battery energy storage system (BESS) and heating/cooling (HVAC) systems.
- **Cooling-dominated Texas climate** with realistic building dynamics, occupant behavior, and weather conditions.
- **Hourly building, weather, and forecast data** for the month of June, enabling predictive and learning-based control.



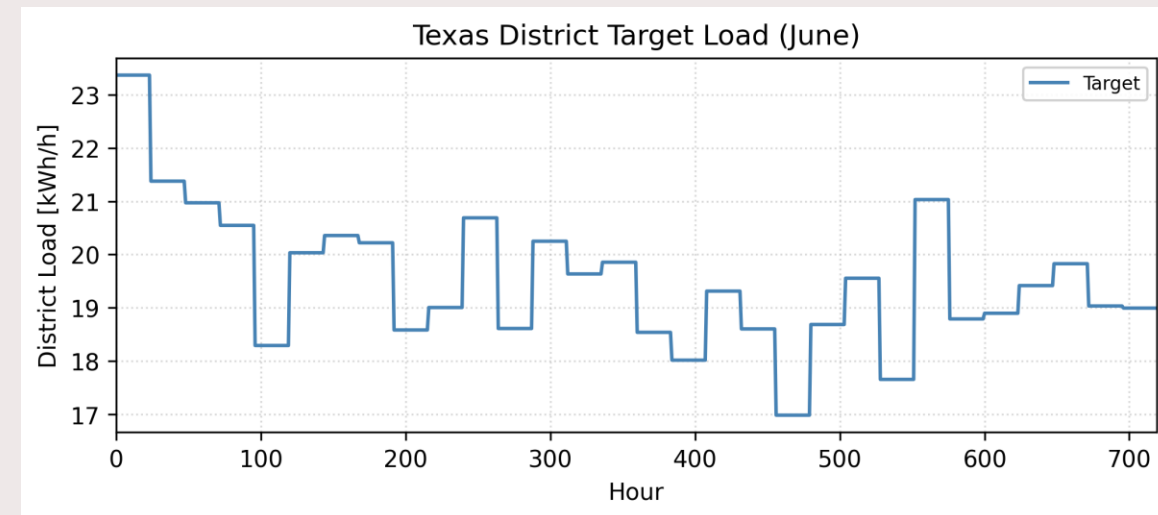
Task We Want to Solve? : Track a Daily Target Load

- **Primary Objective:** Build Controllers that make the total electricity consumption of all 25 buildings follow a target load profile provided by the grid operator as accurately as possible.
- **Secondary Objective:** Maintain indoor temperature to remain within the comfort band, 22–26 °C.
- **Performance Evaluation Metrics:**

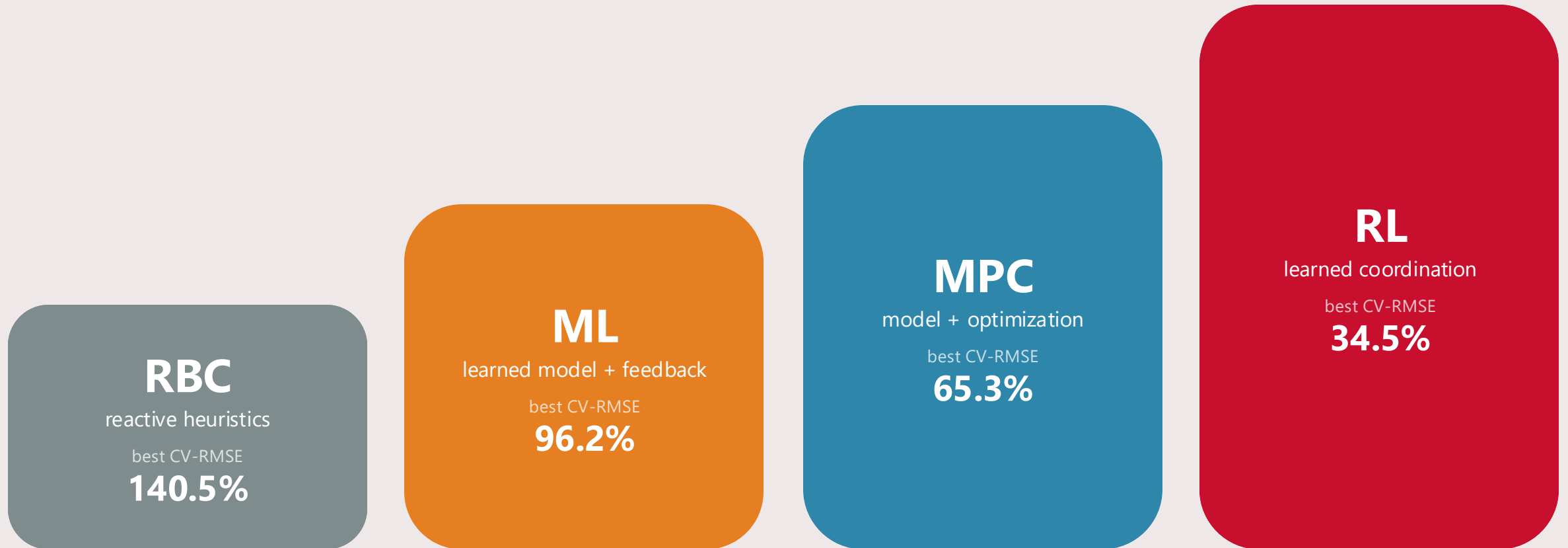
CV-RMSE (%): Measures how accurately the controller follows the target load.

NMBE (%): Tells us whether the controller tends to use more or less energy than the target on average.

Comfort Violation (%): Measures how often indoor temperatures fall outside the desired comfort range.



Roadmap: Four Controller Families, One Ladder



Rule Based Controllers (RBC): What Buildings Use Today

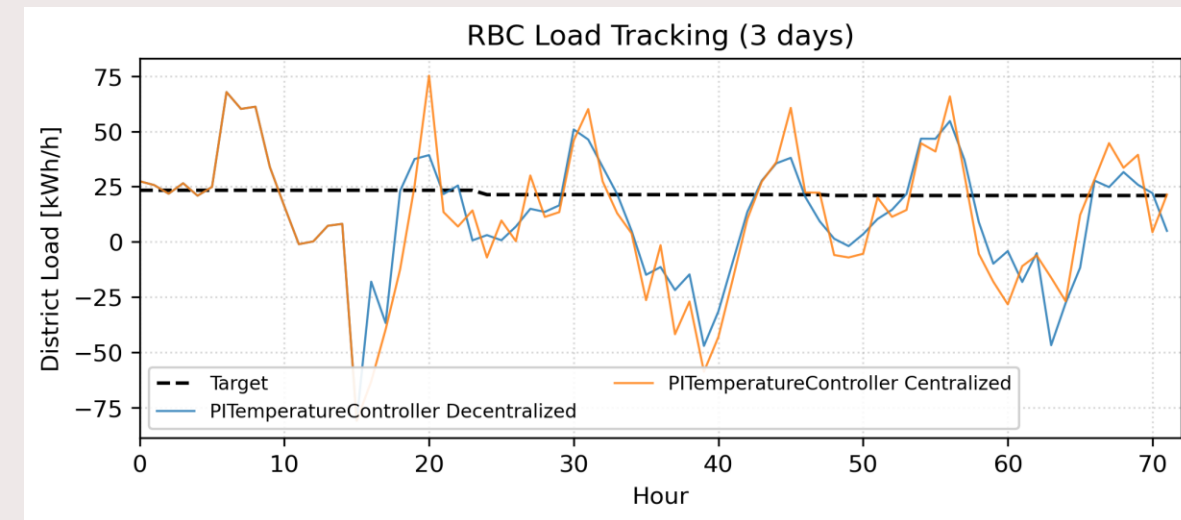
- Controllers that use predefined rules instead of learning or optimization.
- Served as baseline methods for comparison with other approaches.

Controller	Strategy
HourRBC	Uses a fixed 24-hour schedule where actions depend on the current hour of the day.
BasicRBC	Charges the battery at night, discharges during the day, and adjusts cooling intensity based on daytime/nighttime periods.
BasicBatteryRBC	Charges the battery during solar production hours and discharges during the remaining hours to utilize PV generation.
OptimizedRBC	Uses a grid-search optimized daily battery charging and discharging schedule.
TemperatureBasedRBC	Adjusts cooling actions according to the difference between the indoor temperature and the desired comfort temperature.
PITemperatureController	Extends TemperatureBasedRBC by using proportional-integral (PI) feedback to account for both current and accumulated temperature errors.

Results: Rule Based Controllers (RBC)

Decentralized			
Controller	CV-RMSE [%]	NMBE [%]	Comfort Violation [%]
BasicRBC	261.079	166.155	91.39
TemperatureBasedRBC	245.298	-63.133	53.89
OptimizedRBC	220.800	132.824	82.11
BasicBatteryRBC	207.634	89.575	62.14
HourRBC	207.098	115.476	81.90
PITemperatureController	140.523	-54.146	43.64
Centralized			
Controller	CV-RMSE [%]	NMBE [%]	Comfort Violation [%]
TemperatureBasedRBC	266.110	-82.766	82.80
BasicRBC	261.079	166.155	91.39
OptimizedRBC	220.800	132.824	82.11
BasicBatteryRBC	207.634	89.575	62.14
HourRBC	206.770	113.729	81.23
PITemperatureController	158.218	-63.078	63.66

- **Best RBC:** In both Centralized and decentralized cases **PITemperatureController** has performed the best in the all 3 metrics.



Model Predictive Controllers (MPC)

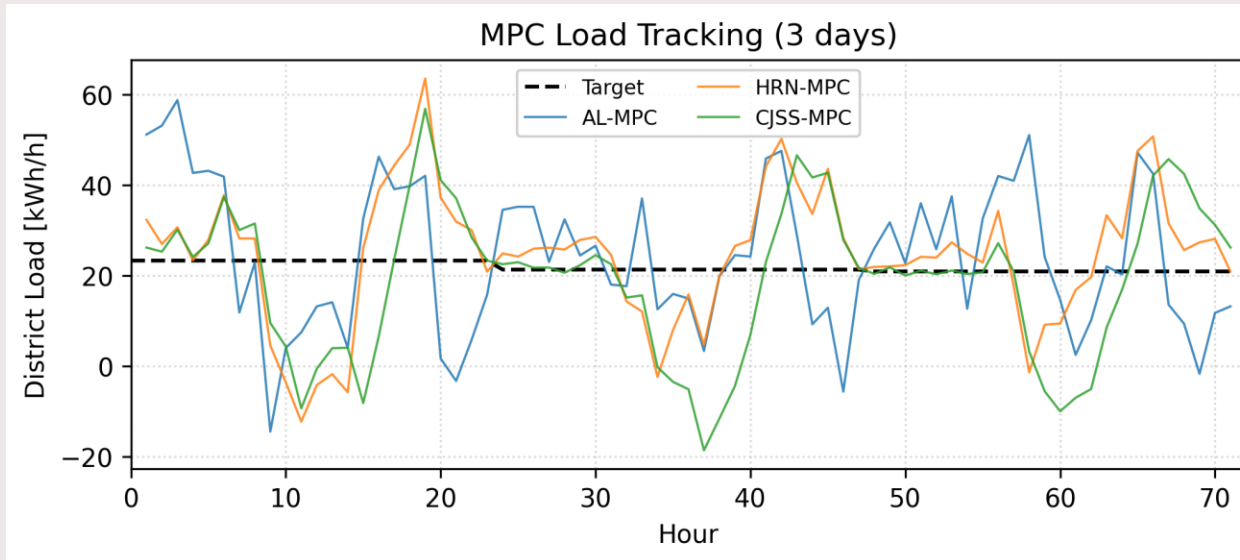
- Controllers that use predictive models and optimization to determine future control actions.

Controller	System Identification Modelling	MPC Features
Adaptive LSTM-Linearized MPC (AL-MPC)	Uses the pre-trained CityLearn LSTM dynamics model and performs local linearization around the current operating point.	Decentralized MPC with distributed district targets; jointly optimizes load tracking, comfort, smoothness, and battery SoC.
Hybrid Residual-NARX MPC (HRN-MPC)	Hybrid model combining linear regression with a residual neural network to capture nonlinear thermal dynamics.	Data-driven predictive MPC using indoor temperature forecasts; optimizes tracking accuracy, comfort, and control effort.
Coordinated Joint State-Space MPC (CJSS-MPC)	Capturing thermal, electrical, and battery dynamics simultaneously.	Hierarchical coordination layer allocates district targets based on building flexibility and enables district-level cooperation.

Results: Model Predictive Controllers (MPC)

Controller	CV-RMSE [%]	NMBE [%]	Comfort Violation [%]
AL-MPC	79.643	10.979	31.42
CJSS-MPC	74.717	-8.083	33.81
HRN-MPC	65.319	18.478	15.33

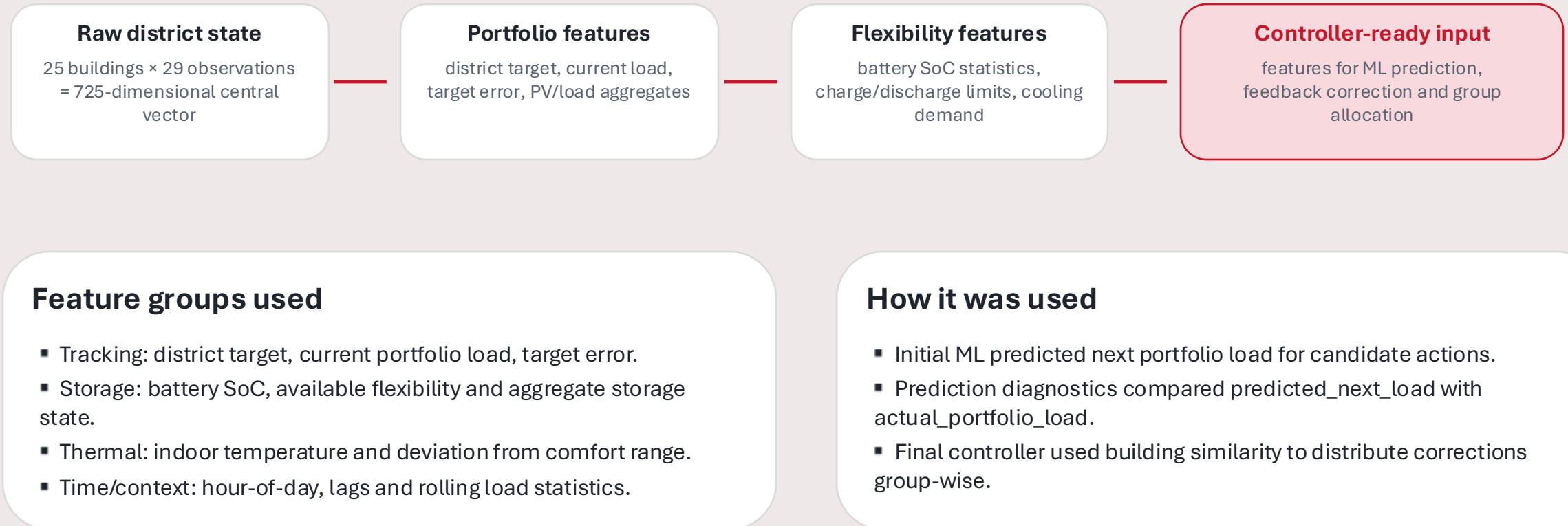
- **Best MPC:** HRN-MPC achieved the best overall MPC performance, providing the strongest balance between the primary (CV-RMSE) and secondary objectives (Comfort Violation).



- The positive NMBE indicates that HRN-MPC tends to consume slightly more energy than the target load on average.

Feature engineering: from raw states to control inputs

converted raw CityLearn observations into compact features for prediction, diagnosis and group-aware control.



Centralized target-aware controller

A centralized baseline that reacts directly to the mismatch between the district target and current load.

What it is

- One controller observes the full district.
- Uses target – current portfolio load.
- Reactive, not predictive, and not ML.

177.0%

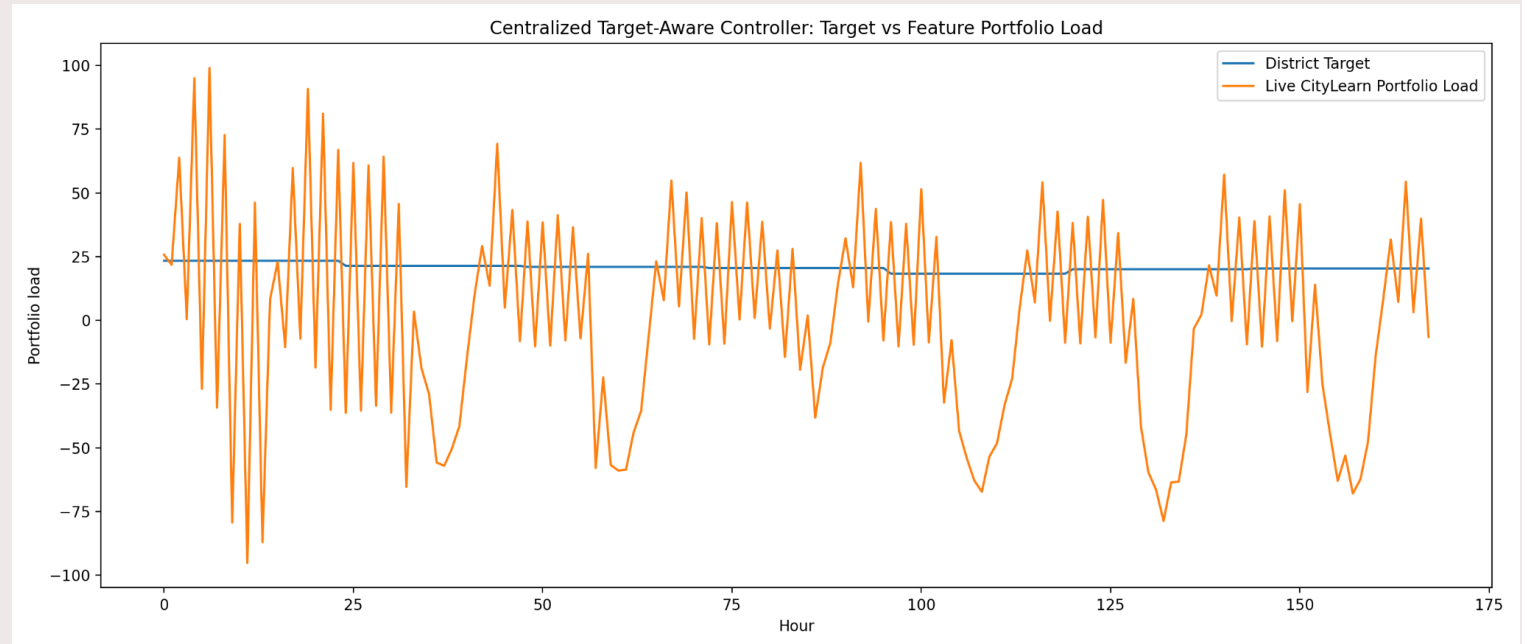
CV-RMSE

-17.8%

NMBE

Why we include it

It shows the limitation of using current tracking error alone: the portfolio load oscillates and does not reliably follow the target.



Observation: large oscillations and negative dips indicate weak closed-loop tracking.

ML predictive controller

The controller searched over candidate battery/cooling actions and chose the action closest to the target in prediction.

Methodology

- Model: HistGradientBoostingRegressor.
- Predicts next portfolio load.
- Action search over battery and cooling commands.
- Selects minimum predicted target error.

146.4%

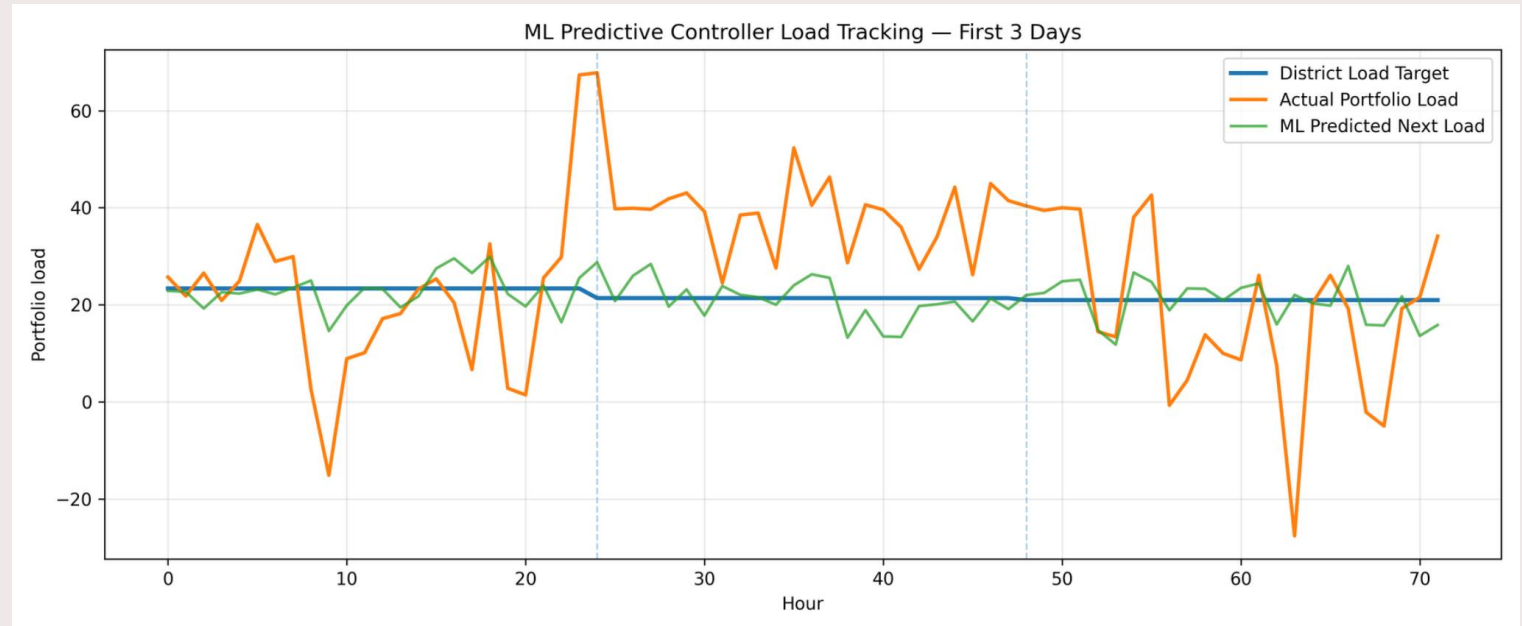
CV-RMSE

+67.8%

NMBE

Diagnosis

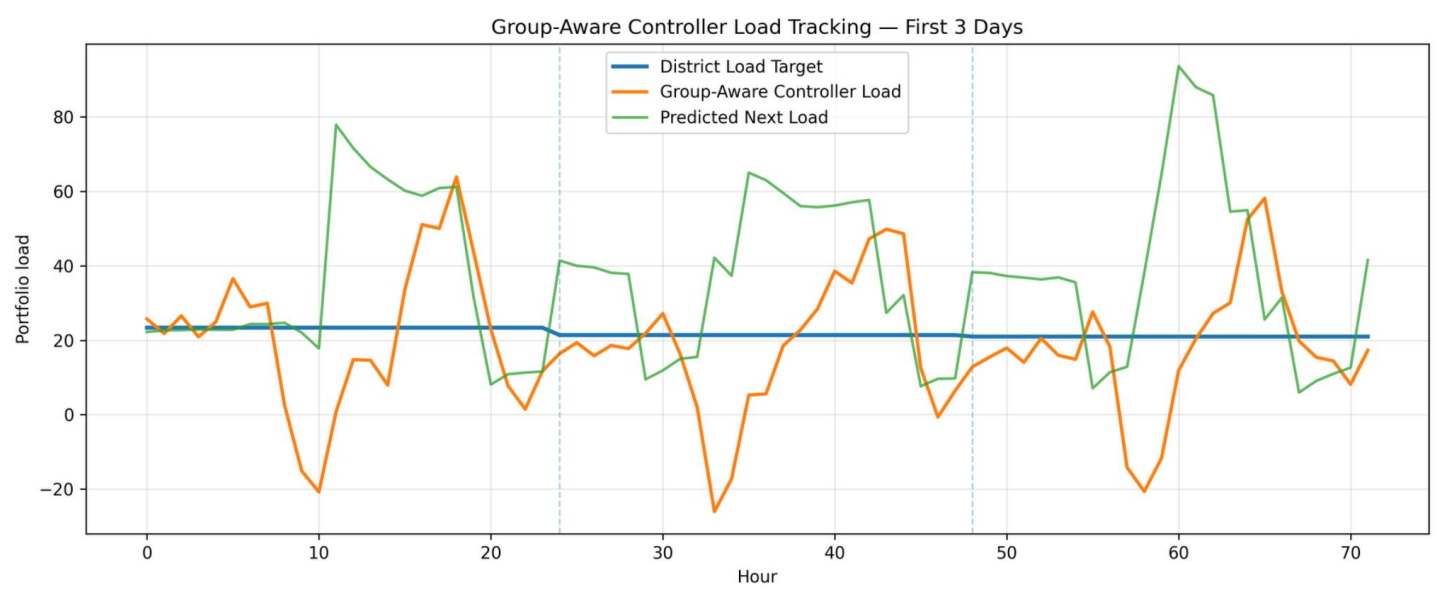
The model prediction stayed near the target, but the actual post-action load diverged after CityLearn dynamics.



Key issue: prediction–realization gap and closed-loop distribution shift.

Final group-aware ML controller

The final design combines ML prediction, feedback correction and group-wise action allocation.



Final controller better aligns realized portfolio load with the target across the 3-day window.

Methodology

- ML proposal: predict next portfolio load for each candidate battery/cooling action.
- Feedback correction: use current tracking error to reduce persistent NMBE bias.
- Group allocation: distribute correction across similar building clusters using SoC, cooling demand and flexibility.

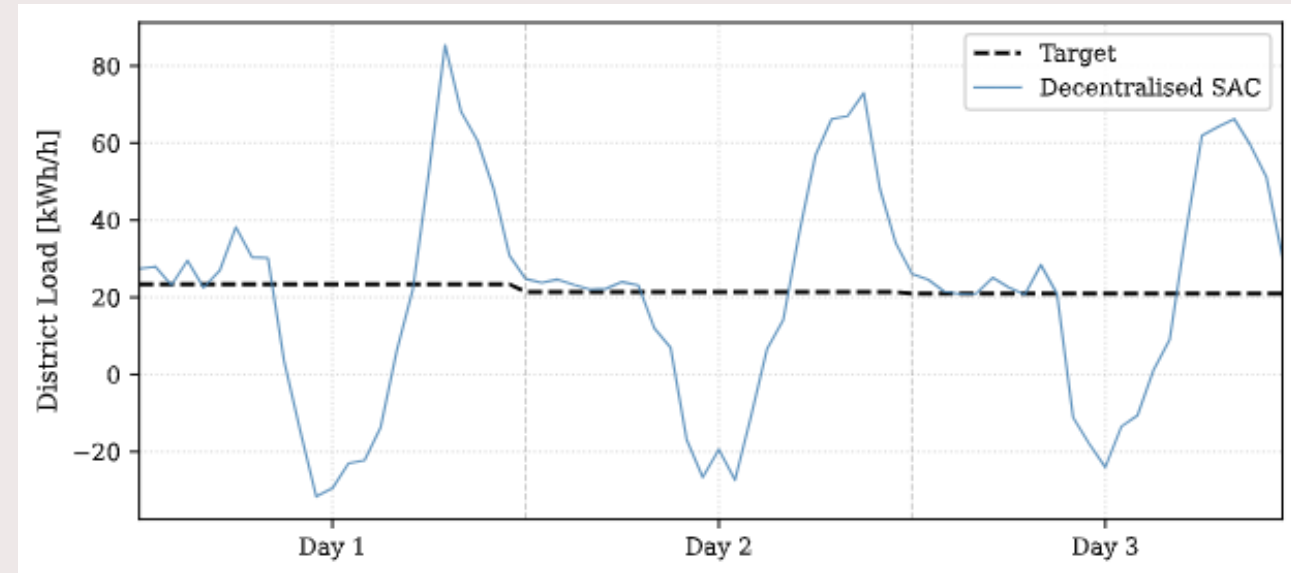
Controller	CV-RMSE	NMBE
Centralized target-aware	177.0%	-17.8%
Initial ML predictive	146.4%	+67.8%
Final group-aware ML	96.2%	≈ -11%

Final message

CV-RMSE improved by about 34% versus the initial ML controller; NMBE moved closer to zero.

Decentralized SAC: right mean, wrong hours

- 25 independent agents, battery-only, 9-dim state, shared tracking reward
- **Soft Actor-Critic** – Learns from past experiences, slightly different actions
- Perfect bias, but great variance, poor coordination



0.65%

NMBE — best of study

134.4%

CV-RMSE

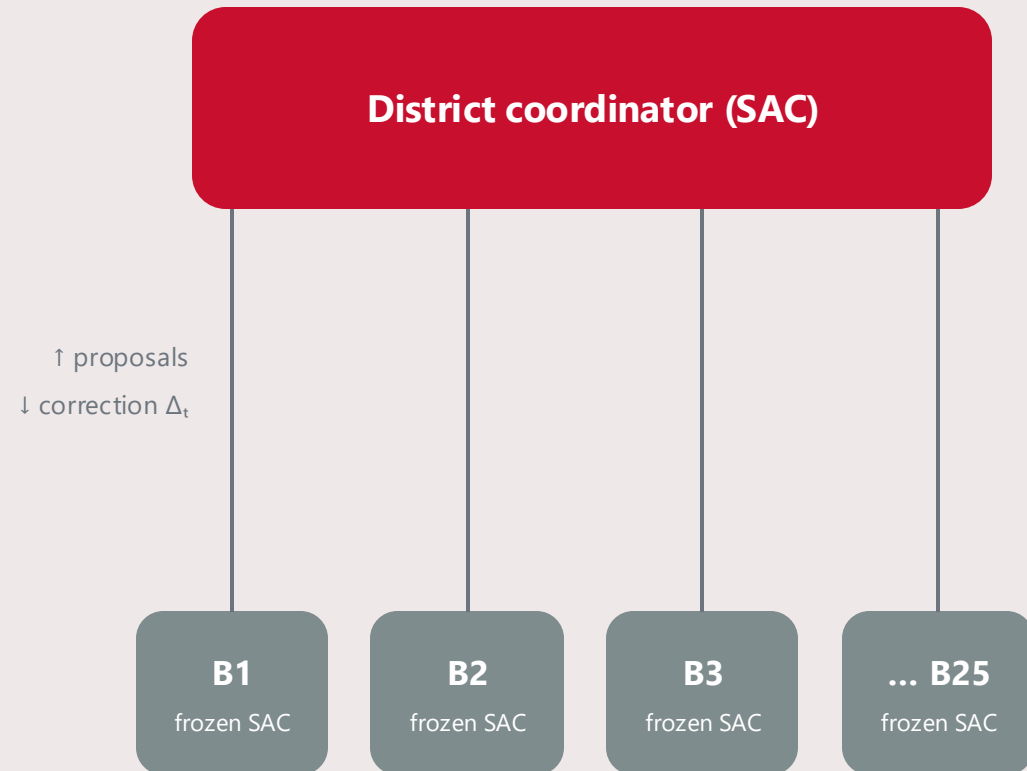
$$J(\pi) = \sum_t E[r(s_t, a_t) + \alpha \cdot H(\pi(\cdot | s_t))]$$

100 episodes × 720 h · ≈9.4 h CPU

Hier-2: one coordinator, 25 frozen specialists

- Coordinator sees 6 h / 12 h ahead targets
→ pre-positions batteries
- EMA-smoothed scalar correction added to all proposals
- Action penalty protects the frozen lower-level policies

$$\Delta_t = 0.4 \cdot r_{\text{raw}} + 0.6 \cdot \Delta_{t-1} \quad (\text{scale } 0.35)$$
$$r_t = -0.01(L_t - d^*_t)^2 - 0.05 \cdot \Delta_t^2$$



Hier-2: best in class

- Beats best MPC
- Five ablations (solar, longer forecasts, SoC stats, larger net, asymmetric scaling): none improved
- Driver = hierarchy + multi-step forecasts

34.5%

CV-RMSE

-74%

Vs. decentralized SAC

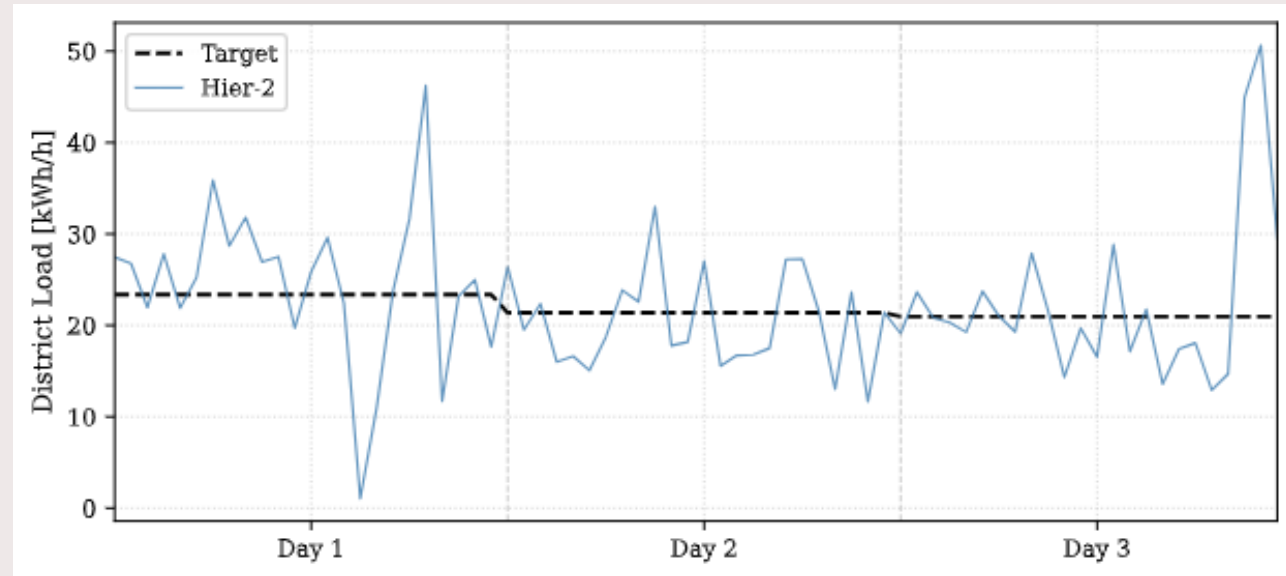
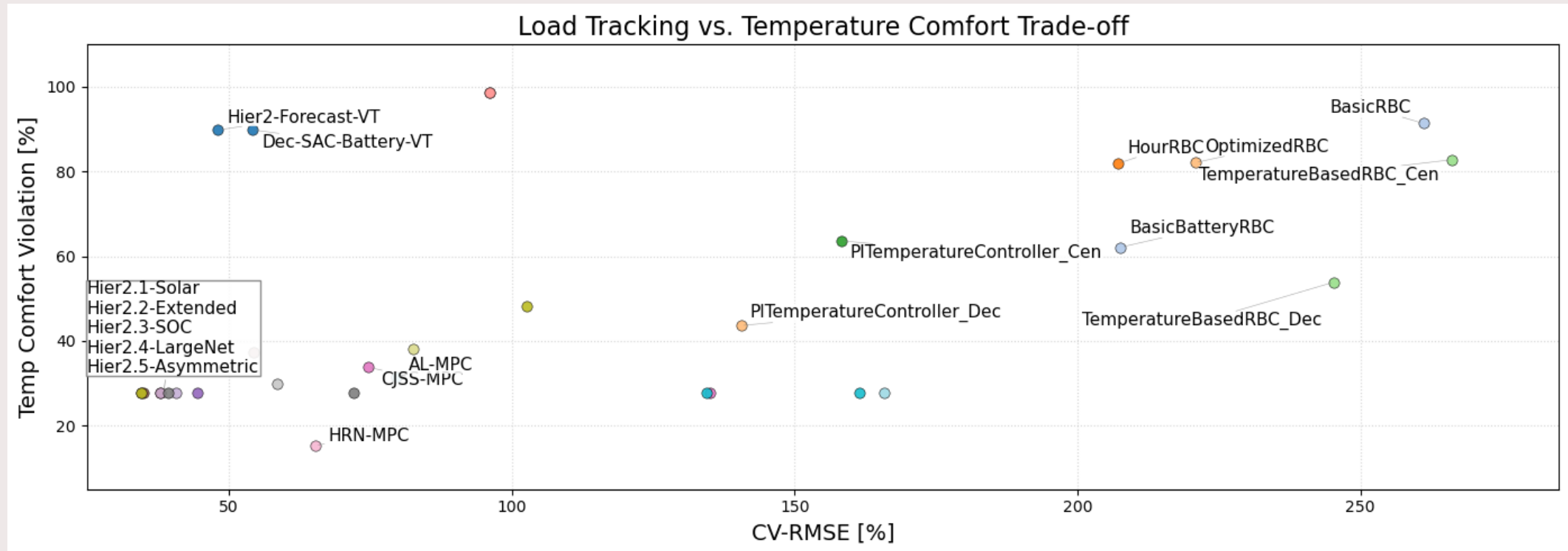


TABLE VI
KPI COMPARISON: HIER-2 VARIANTS VS. BASELINE

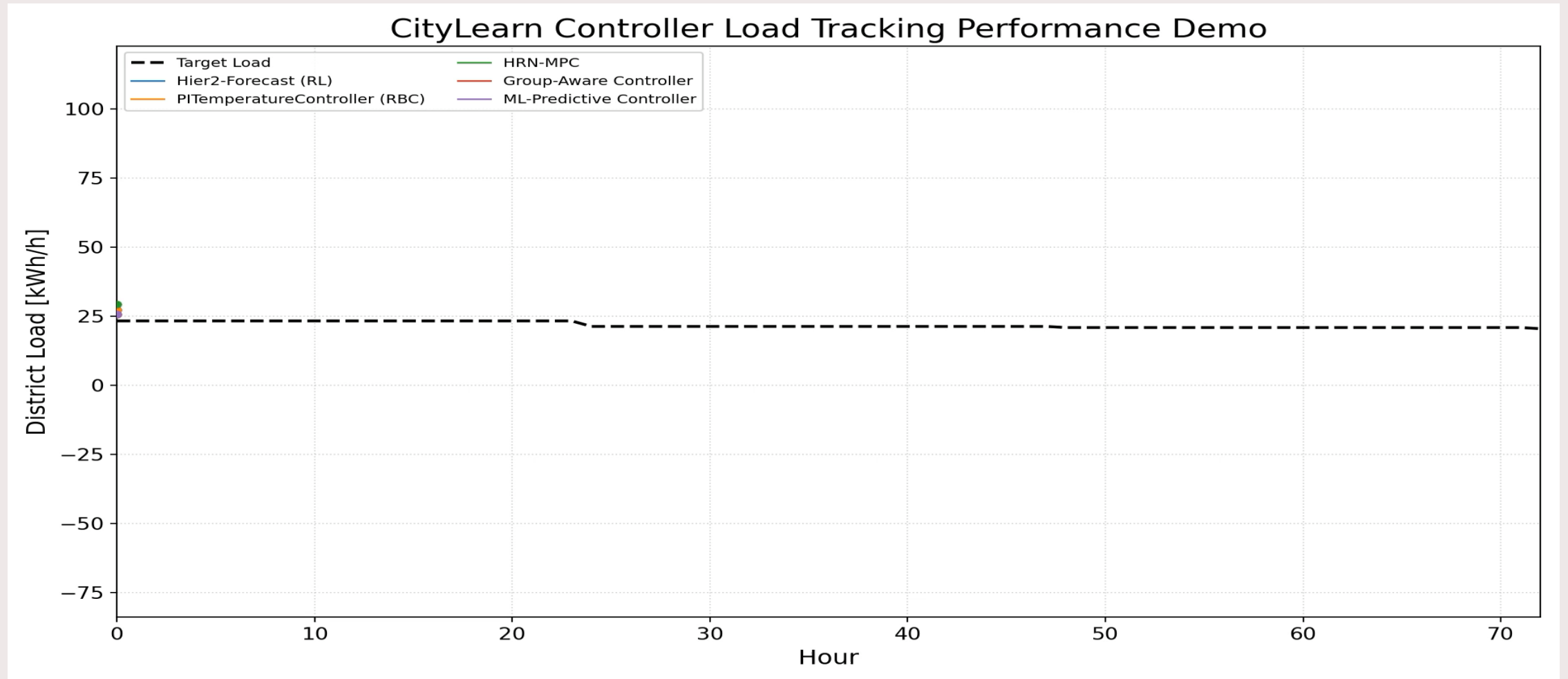
Controller	CV-RMSE [%]	NMBE [%]	Comfort Violation [%]
Hier-2 (baseline)	34.5	5.08	27.8
Hier-2.3-SOC	34.5	5.25	27.8
Hier-2.1-Solar	35.0	4.77	27.8
Hier-2.5-Asymmetric	38.0	5.85	27.8
Hier-2.2-Extended	40.7	5.07	27.8
Hier-2.4-LargeNet	44.5	4.69	27.8

The Big Picture: 3 Clean Clusters

- **RBC** → Lowest Overall Performance
- **MPC / ML** → Intermediate Performance, HRN-MPC Best Thermal Comfort (15.3%)
- **Hier-2 (RL)** → Best Load Tracking (CV-RMSE = 34.5%)
- **Down-Left** → More Prediction, Adaptation and Coordination



Live Demonstration



Thank you — Questions?

Scan the QR code to view the codebase, implementation details, and project report.



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